

Hamiltonians and Nonlocal Games

AlgoLab Summer School — Lecture 2

Adam Bene Watts

Prelude: Why Think About Things Other Than Circuits?

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 - **Cross-pollination.** Techniques developed to solve other quantum problems can be ported back to yield **quantum computational advantage**.
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 - **Cross-pollination.** Techniques developed to solve other quantum problems can be ported back to yield **quantum computational advantage**.
~> *We will see an explicit example of this later in the lecture!*
 - **Intuition.** Studying other quantum problems sharpens our intuition for **how quantum mechanics behaves**.

Part 1: Classical Constraint Satisfaction Problems

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- The **locality** of a CSP is the largest k in any constraint. We focus on **constant locality**: k fixed as n grows.

Example 1: 3XOR

- In a 3XOR instance, each constraint checks the parity of exactly 3 variables:

$$C_{ijk}(X_i, X_j, X_k) = \begin{cases} 1 & \text{if } X_i \oplus X_j \oplus X_k = b_{ijk} \\ 0 & \text{otherwise} \end{cases}$$

for some target bit $b_{ijk} \in \{0, 1\}$.

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- Q:** How hard is it to determine whether all constraints in a 3XOR problem can be satisfied?
- A:** **Easy!** (See next slide.)

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- Parity constraints are **linear equations over $\mathbb{F}_2$** . Encode the system as $Ax = b$:

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- A:** **Hard!** (NP-hard — to be defined this afternoon.)
- In fact, an even simpler problem is still NP-hard.

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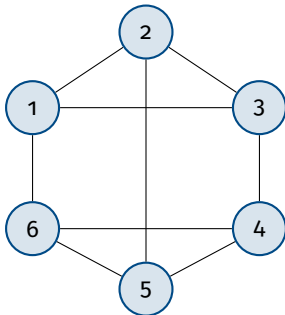
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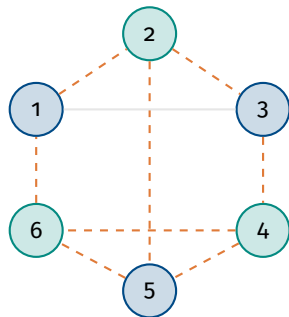
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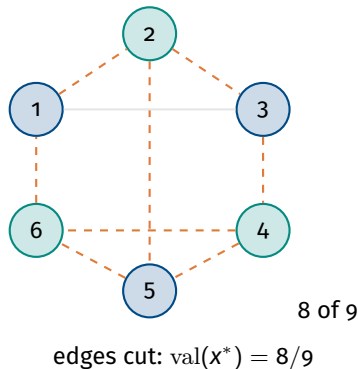


8 of 9

edges cut: $\text{val}(x^*) = 8/9$

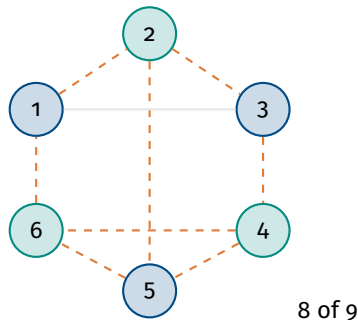
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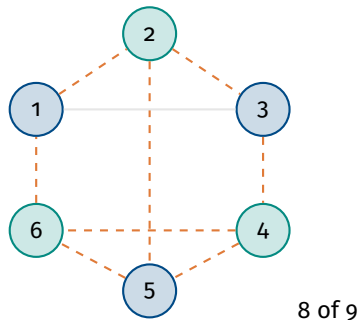
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- Random assignments achieve $\text{val} = 1/2$ in expectation.

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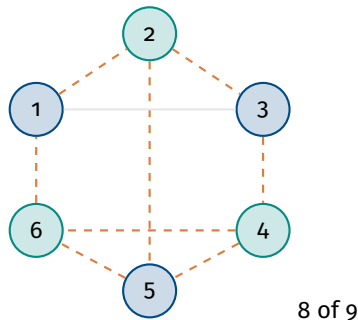
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- **Goemans-Williamson:** a polynomial-time algorithm achieving at least 0.878 of the optimum.
- The **Unique Games Conjecture** implies it is NP-hard to do better.

Part 2: The Local Hamiltonian Problem

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k -Local Hamiltonian Problem. Given Hermitian matrices $H_1, \dots, H_m$ each acting nontrivially on at most k qubits of an n -qubit system, find the state $|\psi\rangle$ minimising

$$\langle \psi | H | \psi \rangle, \quad \text{where} \quad H = \frac{1}{m} \sum_{i=1}^m H_i.$$

Example: Quantum Max Cut

- Given a graph $G = (V, E)$, the **Quantum Max Cut** (QMC) Hamiltonian is:

$$H_{\text{QMC}} = \frac{1}{|E|} \sum_{(i,j) \in E} H_{ij}, \quad H_{ij} = \frac{1}{4} (I - XX - YY - ZZ)_{ij}$$

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Exercise. Show that if we kept only the $Z_i Z_j$ and identity terms in the QMC Hamiltonian, we would recover classical Max Cut.

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Question. What is the ground state (minimum energy state) of the two-qubit QMC Hamiltonian

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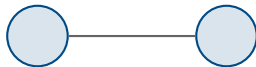
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The eigenvalues of SWAP are +1 (symmetric) and -1 (antisymmetric), so H has eigenvalues 0 and 1. The ground state energy is 0, achieved by the **singlet** $|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$.

QMC on More Qubits and Monogamy of Entanglement

Question. Is there a state that achieves energy 0 on the three QMC Hamiltonian associated with a three vertex line?
Equivalently, does there exist a $|\psi\rangle$ on three qubits such that

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In general, solving QMC is believed to be **hard even for a quantum computer**. Verifying the energy of a candidate state is easy quantumly but hard classically. Formally: **QMA-hard** (definitions this afternoon!).

- **Problem definition and first approximation algorithm**

S. Gharibian and O. Parekh, *Almost Optimal Classical Approximation Algorithms for a Quantum Generalization of Max-Cut*, APPROX/RANDOM 2019. arXiv:1909.08846.

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- **Integrality gaps and hardness of approximation**

Y. Hwang, J. Neeman, O. Parekh, K. Thompson, J. Wright, *Unique Games Hardness of Quantum Max-Cut, and a Conjectured Vector-Valued Borell's Inequality*, SODA 2023. arXiv:2111.01254.

References and Further Reading: Quantum Max Cut

- **Problem definition and first approximation algorithm**

S. Gharibian and O. Parekh, *Almost Optimal Classical Approximation Algorithms for a Quantum Generalization of Max-Cut*, APPROX/RANDOM 2019. arXiv:1909.08846.

- **Improved approximation algorithms**

E. Lee and O. Parekh, *An Improved Quantum Max Cut Approximation via Maximum Matching*, ICALP 2024. arXiv:2401.03616.

- **Integrality gaps and hardness of approximation**

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- **Algebraic structure of QMC**

A. Bene Watts, A. Chowdhury, A. Epperly, J.W. Helton, I. Klep, *Relaxations and Exact Solutions to Quantum Max Cut via the Algebraic Structure of Swap Operators*, Quantum 8, 1352 (2024). arXiv:2307.15661.

Part 3: Nonlocal Games

Testing CSPs

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- **Q:** How could we be convinced of the value of a k -local CSP? For concreteness, consider this 3XOR instance:

$$A_0 \oplus B_0 \oplus C_0 = 0$$

$$A_1 \oplus B_1 \oplus C_0 = 1$$

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Q: How could we test the maximum fraction of clauses satisfiable in:

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- Check if the XOR of responses matches the target bit. If so, players **win**; otherwise they **lose**.

Soundness of the Three-Prover Test

Claim. The maximum winning probability of the three players equals the value of the 3XOR CSP.

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Proof. Let $\tilde{A}_i$ be player 1's response to question i ; define $\tilde{B}_i, \tilde{C}_i$ similarly. Responses may depend on shared randomness $\lambda \sim p(\lambda)$. The average win probability is:

$$\frac{1}{4} \sum_{\lambda} p(\lambda) \left[\Pr \left[\tilde{A}_0(\lambda) \oplus \tilde{B}_0(\lambda) \oplus \tilde{C}_0(\lambda) = 0 \right] + \dots \right. \\ \left. + \Pr \left[\tilde{A}_0(\lambda) \oplus \tilde{B}_1(\lambda) \oplus \tilde{C}_1(\lambda) = 1 \right] \right]$$

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Picking the λ^* that maximizes the bracketed expression gives a deterministic CSP assignment achieving at least the average win probability. □

Quantum Generalization: The GHZ Game

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The GHZ Game.

- The verifier picks a clause uniformly at random from the four clauses of our 3XOR instance.
- Each prover receives the index of their variable (0 or 1) in that clause, and responds with a bit.
- The provers **win** if their responses satisfy the clause.
- Before the game, provers may prepare any shared quantum state. During the game they **cannot communicate**.

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Question. What is the maximum winning probability for the provers in the GHZ game?

Strategy for the GHZ Game

GHZ Strategy. Before the game, the provers prepare:

$$|\text{GHZ}\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

each holding one qubit. On receiving question $q \in \{0, 1\}$:

$q = 0$: measure in the X basis $\{|+\rangle, |-\rangle\}$

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Respond with 0 for + outcome, 1 for – outcome.

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Respond with 0 for + outcome, 1 for – outcome.

Claim: this strategy wins with probability 1!

Strategy for the GHZ Game: Analysis

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So $|\text{GHZ}\rangle$ is a $+1$ eigenstate of XXX and a -1 eigenstate of XYY , YXY , YYX .

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Measuring XXX always gives an **even** number of – outcomes; measuring XYY , YXY , or YYX always gives an **odd** number. Their responses always have the winning parity. □

Part 3: Circuits and Nonlocal Games

Circuit Separations

- Recall from Lecture 1: circuits of 1- and 2-qubit gates can implement **any unitary**. But physical qubits are subject to **noise** — the more depth, the more noise accumulates.

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- **Weaker Q:** Can shallow quantum circuits achieve quantum advantage over shallow *classical* circuits?
A: **Yes!**

Theorem (Bravyi–Gosset–König, 2018). There exists a computational problem solvable by a constant-depth quantum circuit that **cannot** be solved by any constant-depth classical circuit.

Quantum Advantage with Shallow Circuits 1

Theorem. QNC^0 circuits (constant-depth, 1- and 2-qubit gates) can solve a relational problem (mapping input bits to output bits) that NC^0 circuits (constant-depth, unbounded fan-out, AND/OR gates) cannot.

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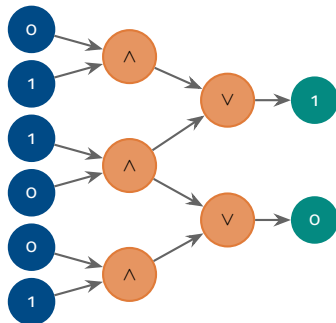
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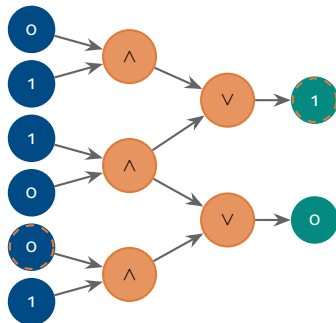


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- In an NC^0 circuit, each output bit depends on only a **constant number** of input bits.
- So an NC^0 circuit **cannot** win the GHZ game: some player's output is independent of another player's question, breaking the required correlations.



Quantum Advantage with Shallow Circuits 2

Proof sketch, Part 2: Quantum upper bound.

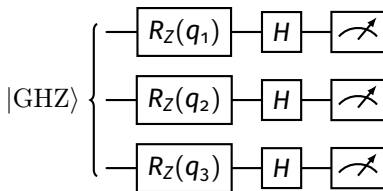
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- Players share a **GHZ state**. Each receives a question bit q_i , applies a classically-controlled R_z rotation, then measures in the X basis — implementing the GHZ strategy in constant depth.

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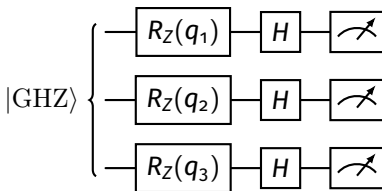
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Problem. A GHZ state spanning n qubits cannot be prepared by a QNC^0 circuit — its preparation requires depth at least $\Omega(\log n)$.

Quantum Advantage with Shallow Circuits: Cluster State

Proof sketch, Part 3: Fix with cluster states.

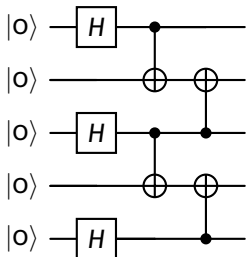
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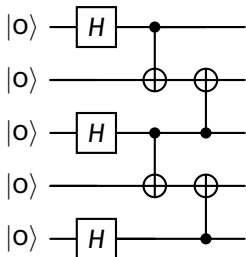
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- Preparing this state and making GHZ measurements controlled on inputs produces **non-classical** correlations. □

References and Further Reading: Circuit Separations

- **Original circuit separation**

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- **Experimental implementation (trapped-ion)**

A. K. Daniel, Y. Zhu, C. H. Alderete et al., *Quantum Computational Advantage Attested by Nonlocal Games with the Cyclic Cluster State*, Phys. Rev. Research 4, 033068 (2022). arXiv:2110.04277.